

Bodhisattwa Dhara

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EDUCATION

BITS Pilani, Hyderabad Campus

Minor in Data Science

Hyderabad, Telangana

2021 – 2025

- Academic focus included **Data Science, Machine Learning, Statistical Modelling, Deep Learning, Natural Language Processing**, and applied AI research.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

BITS Pilani, Hyderabad Campus

Aug 2023 – May 2025

Hyderabad, Telangana

- Conducted research on **Machine Learning Enabled Fluidic Droplet Generation** under the SPARKLE research program, combining experimental systems with data-driven modelling.
- Applied **K-Nearest Neighbours, Support Vector Regression, Random Forests** and time-series analysis to model and predict fluid-flow behaviour from experimental parameters.
- Worked at the intersection of **scientific machine learning, experimental modelling and engineering**, developing model explanations, research documentation and technical presentations.
- Developed broader research exposure in **clinical LLM vulnerabilities, data poisoning, healthcare NLP and trustworthy AI**, contributing to peer-reviewed and preprint research outputs.

Research Intern – Healthcare AI

IIT Guwahati

May 2024 – Jul 2024

Guwahati, Assam

- Conducted research on **TinyML-enabled healthcare AI systems**, focusing on resource-efficient ECG analysis and intelligent diagnostic-support systems.
- Explored the integration of **ECG and healthcare signal data with Large Language Models and Retrieval-Augmented Generation**, with emphasis on contextual clinical reasoning.
- Investigated diagnostic context retrieval, explainability, resource-efficient inference and deployment-aware AI system design for healthcare applications.

Algorithm Analysis Research Intern

CSIR-CEERI Pilani

May 2023 – Jul 2023

Pilani, Rajasthan

- Worked on **real-time algorithm prediction for low-powered IoT devices**, focusing on embedded sensor systems, resource-constrained predictive modelling and edge deployment constraints.
- Studied **PPG-based blood-pressure and physiological monitoring systems** and optimised memory usage for sensor-data processing, improving system efficiency and prediction quality by **20%**.
- Used regression-based machine-learning models for temperature prediction and sensor-behaviour analysis across low-power sensor environments.

PUBLICATIONS & RESEARCH MANUSCRIPTS

A. Das, A. Tariq, F. Batalini, B. Dhara, and I. Banerjee. “Exposing Vulnerabilities in Clinical LLMs Through Data Poisoning Attacks: Case Study in Breast Cancer.” *AMIA Annual Symposium Proceedings*, pp. 339–348. 2024

- Peer-reviewed publication** investigating vulnerabilities of clinical large language models to instruction-based data poisoning and targeted model-editing attacks using breast-cancer clinical data.
- Research focused on **clinical LLM robustness, adversarial machine learning, trustworthy AI and healthcare NLP**.
- [PubMed Indexed Publication](#)

A. Das, A. Tariq, F. Batalini, B. Dhara, and I. Banerjee. “Framework for Exposing Vulnerabilities of Clinical Large Language Model: A Case Study in Breast Cancer.” *TechRxiv Preprint*. 2024

- Extended the clinical LLM vulnerability framework to study **instruction-based data poisoning, trigger-based model editing, membership inference and privacy leakage** in healthcare language models.
- Evaluated risks associated with applying generative AI to sensitive clinical environments, including reliability, model manipulation and patient-data privacy.
- [TechRxiv DOI](#)

WORK EXPERIENCE

Data Scientist / Junior Data Scientist

May 2025 – Present

SG Analytics

Bengaluru, India

- Built **Bayesian hierarchical classification models** in PyMC for enquiry-level prediction, using School to Subregion to Global hierarchies, posterior probabilities, HDI intervals, calibration and feature-reduction methods including VIF, IV, mutual information and PhiK.
- Developed **call-centre forecasting and workforce-planning pipelines** using Prophet, ARIMA, Poisson simulation, Erlang-C queueing and OR-Tools CP-SAT to estimate demand, staffing requirements, rosters, coverage and service-level trade-offs.
- Contributed to an **enterprise PowerPoint automation system** that extracts slide content into JSON, plans slide-level changes and regenerates client-ready decks using Python, Streamlit, PowerPoint automation and LLM/agent orchestration.
- Worked on **GenAI and natural-language analytics workflows** using dbt semantic models, MetricFlow, AWS Bedrock and LIDA-style chart generation to connect governed business metrics with natural-language questions and automated visual explanations.

Data Science Intern

Aug 2024 – Dec 2024

SG Analytics

Bengaluru, India

- Automated parsing and analysis of **PNML-based Petri-net models** using PM4Py and NetworkX, extracting places, transitions, arcs, markings, process paths and structured CSV/JSON outputs.
- Prepared and transformed multi-source event data for process discovery, workflow analysis, bottleneck identification and process visualisation.

TECHNICAL SKILLS

Programming & Data: Python, SQL, C/C++, Git

Machine Learning & Statistics: Scikit-learn, regression, classification, clustering, time-series forecasting, Bayesian hierarchical modelling, feature selection, probability calibration, backtesting

Deep Learning & Healthcare AI: PyTorch, TensorFlow, Keras, TinyML, ECG analytics, clinical NLP, healthcare AI, representation learning

GenAI, NLP & LLM Systems: LLMs, RAG, embeddings, tool calling, chatbot systems, LangGraph-style orchestration, AWS Bedrock, NLP

Data Engineering & BI: Pandas, NumPy, dbt semantic models, MetricFlow, ETL, data validation, process mining, PM4Py, Excel automation

Modelling & Optimisation Tools: PyMC, Prophet, ARIMA, OR-Tools CP-SAT, OpenCV, pdfplumber, Matplotlib, Seaborn, Streamlit

RESEARCH & PUBLICATION HIGHLIGHTS

- **Peer-reviewed and PubMed-indexed publication** in the AMIA Annual Symposium Proceedings on clinical LLM vulnerabilities, data poisoning and trustworthy AI in healthcare.
- Co-authored an extended **TechRxiv research preprint** investigating clinical LLM vulnerabilities through data poisoning, targeted model editing, membership inference and privacy leakage.
- Research exposure spans **healthcare AI, clinical NLP, trustworthy AI, adversarial LLMs, TinyML, IoT sensor analytics, scientific ML, microfluidics, Bayesian modelling and GenAI systems**.
- Research domains: IoT/TinyML sensor analytics, microfluidics ML, ECG diagnosis with LLMs/RAG, process mining, Bayesian modelling and enterprise GenAI automation.